Titanic Survival Prediction using Logistic Regression

Case Study 3

Roll No: 160124733365

Name: B. Madhav

Section: CSE-3

Year: 3rd Year

Subject: Fundamentals of Data Science (FDS)

Methodology

1. **Data exploration:** loaded titanic.csv (891 rows, 12 columns), checked data types, summary statistics and missing values, and plotted count plots, histogram, box plot and correlation heatmap.
2. **Preprocessing:** filled missing Age with median and Embarked with mode, dropped PassengerId, Name, Ticket and Cabin, encoded Sex as 0/1 and one-hot encoded Embarked.
3. **Model building:** 80/20 stratified train-test split, StandardScaler on features, LogisticRegression from scikit-learn.
4. **Evaluation:** accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix and ROC curve on the test set.
5. **Interpretation:** coefficients and odds ratios of each feature.
6. **Deployment:** model saved as model.pkl and served with a Streamlit app (app.py).

Tools: Python, pandas, NumPy, matplotlib, seaborn, scikit-learn, Streamlit. AI assistance (Claude) was used while writing the code.

Case Study 3: Titanic Survival Prediction using Logistic Regression

Roll No: 160124733365 | **Subject:** Fundamentals of Data Science (FDS)

Problem statement: Predict whether a passenger survived the Titanic disaster (Survived = 1) from passenger details such as class, sex, age, family size, fare and port of embarkation, and find which factors most affected survival.

Note: AI assistance (Claude) was used to help write this code; the outputs were run and checked by me.

1. Data Exploration

Importing libraries and loading the dataset.

```
(891, 12)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cab
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	Na
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	Na
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	Na

The dataset has 891 passengers and 12 columns. `Survived` is the target variable.

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

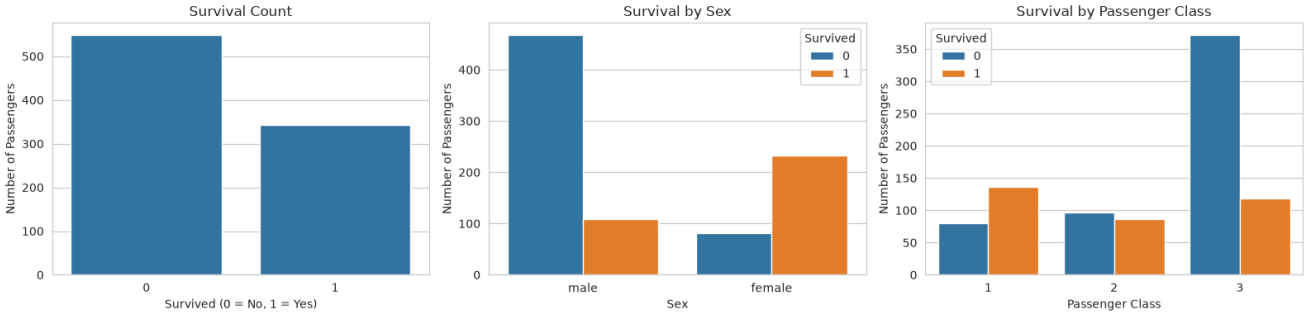
`Age`, `Cabin` and `Embarked` have missing values. `Name`, `Sex`, `Ticket`, `Cabin`, `Embarked` are text (object) columns; the rest are numeric.

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Average age is about 29.7 years. Fare is very skewed (median about 14.5 but max 512). Only about 38% of passengers survived (mean of `Survived`).

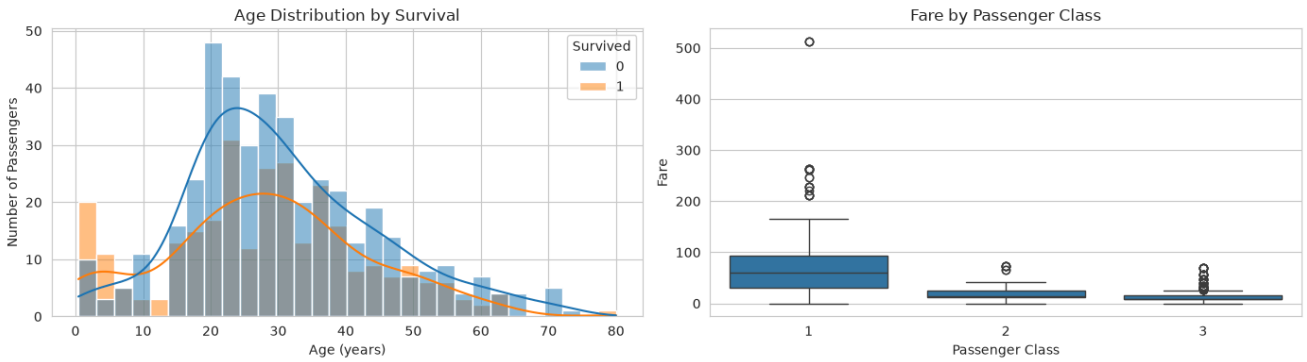
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Cabin is missing for 687 of 891 rows (77%) so it will be dropped. Age (177 missing) and Embarked (2 missing) will be imputed.

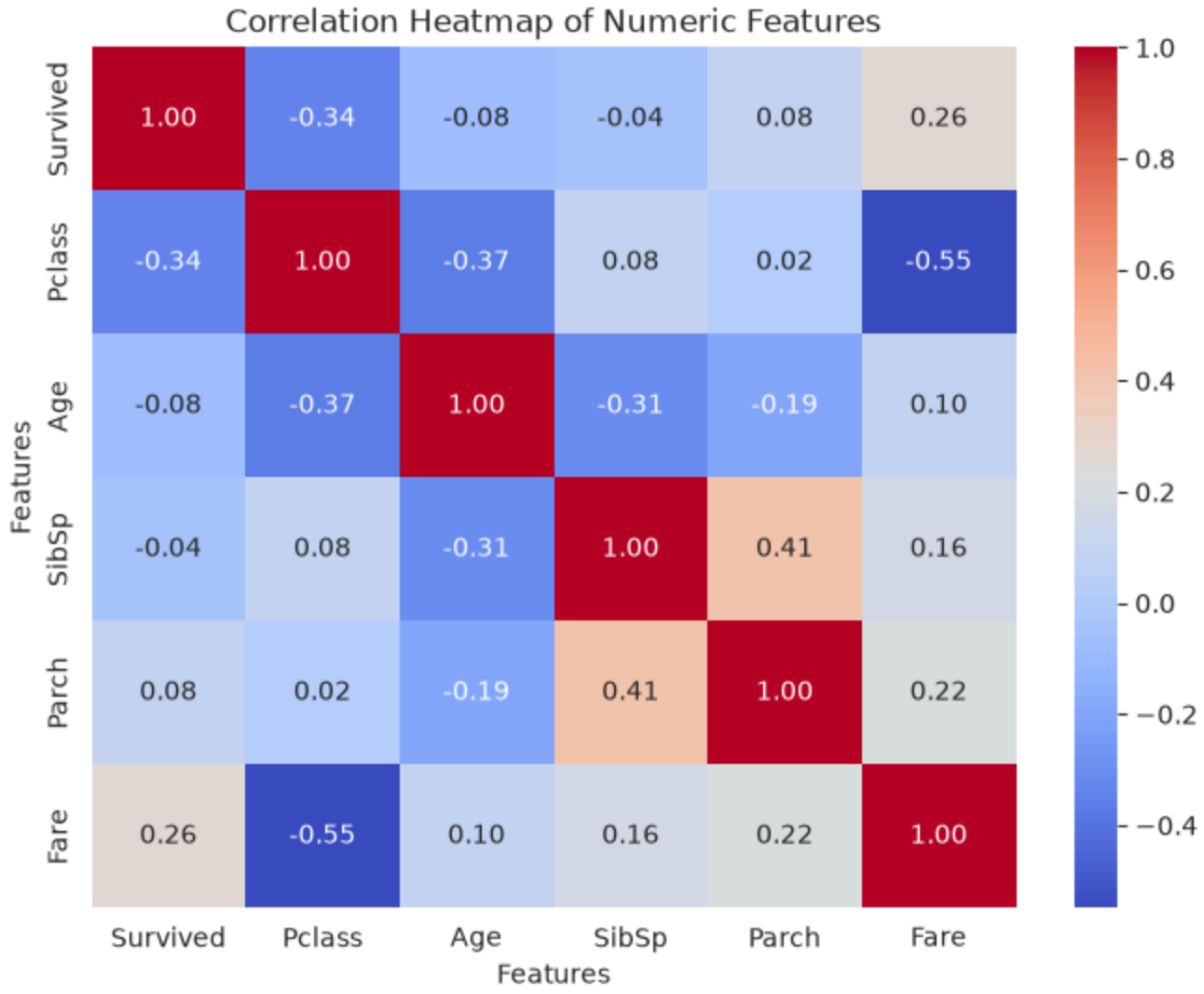



```
Sex
female    0.742038
male      0.188908
Name: Survived, dtype: float64
Pclass
1    0.629630
2    0.472826
3    0.242363
Name: Survived, dtype: float64
```

Interpretation: 549 passengers died and 342 survived, so the classes are a bit imbalanced. About 74% of females survived but only about 19% of males, so sex is a very strong factor. Survival also drops with class: about 63% in 1st class, 47% in 2nd and 24% in 3rd.



Interpretation: Most passengers were 20 to 40 years old. Young children (below about 10) had a better survival rate, which fits the "women and children first" rule. The box plot shows 1st class fares were much higher and had many outliers (up to 512), so fare and class are closely linked.



Interpretation: Pclass has a negative correlation with survival (-0.34): higher class number (3rd class) means lower survival. Fare has a positive correlation (+0.26). Pclass and Fare are strongly negatively correlated (-0.55) with each other. Age, SibSp and Parch have weak direct correlation with survival.

2. Data Preprocessing

0

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked_Q	Embarked_S
0	0	3	0	22.0	1	0	7.2500	0	1
1	1	1	1	38.0	1	0	71.2833	0	0
2	1	3	1	26.0	0	0	7.9250	0	1
3	1	1	1	35.0	1	0	53.1000	0	1
4	0	3	0	35.0	0	0	8.0500	0	1

- Age is filled with the median (28) because age is skewed and median is not affected by outliers.
- Embarked is filled with the mode ('S').
- PassengerId, Name, Ticket are identifiers and Cabin is mostly empty, so they are dropped.
- Sex is encoded as 0/1, and Embarked is one-hot encoded (Embarked_Q, Embarked_S; C is the base).
- After this there are no missing values left.

3. Model Building

Splitting into 80% train / 20% test (stratified so both sets have the same survival ratio) and scaling features so the coefficients can be compared.

```
Train size: (712, 8)  Test size: (179, 8)
Training accuracy: 0.8076
```

The model is trained on 712 passengers and tested on 179 unseen passengers.

4. Model Evaluation

```
Accuracy  : 0.8045
Precision : 0.7931
Recall    : 0.6667
F1-score  : 0.7244
ROC-AUC   : 0.8433

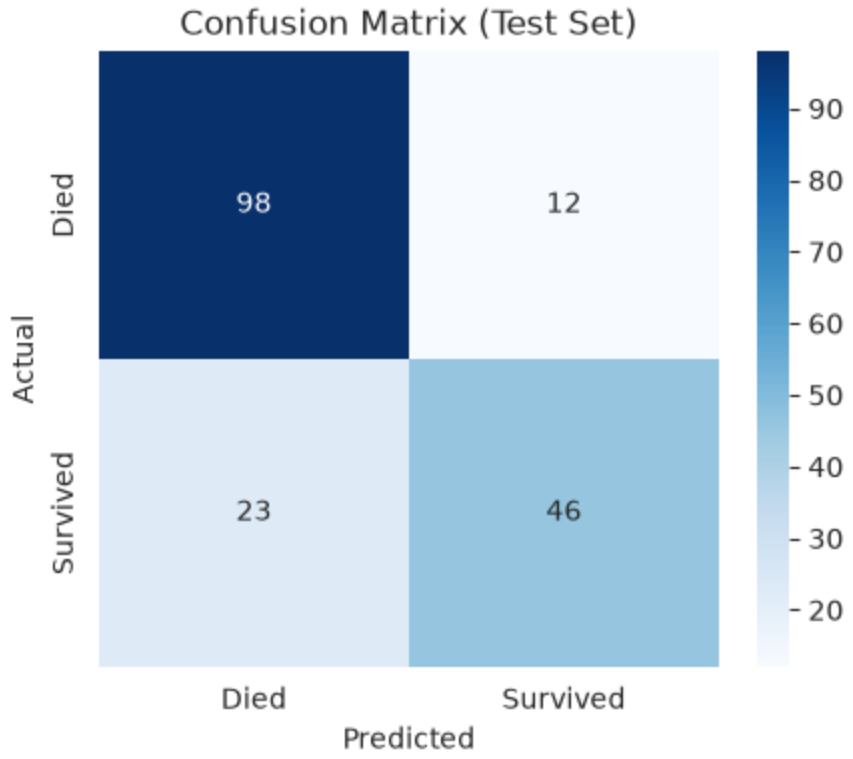
              precision    recall  f1-score   support

           0       0.81      0.89      0.85       110
           1       0.79      0.67      0.72        69

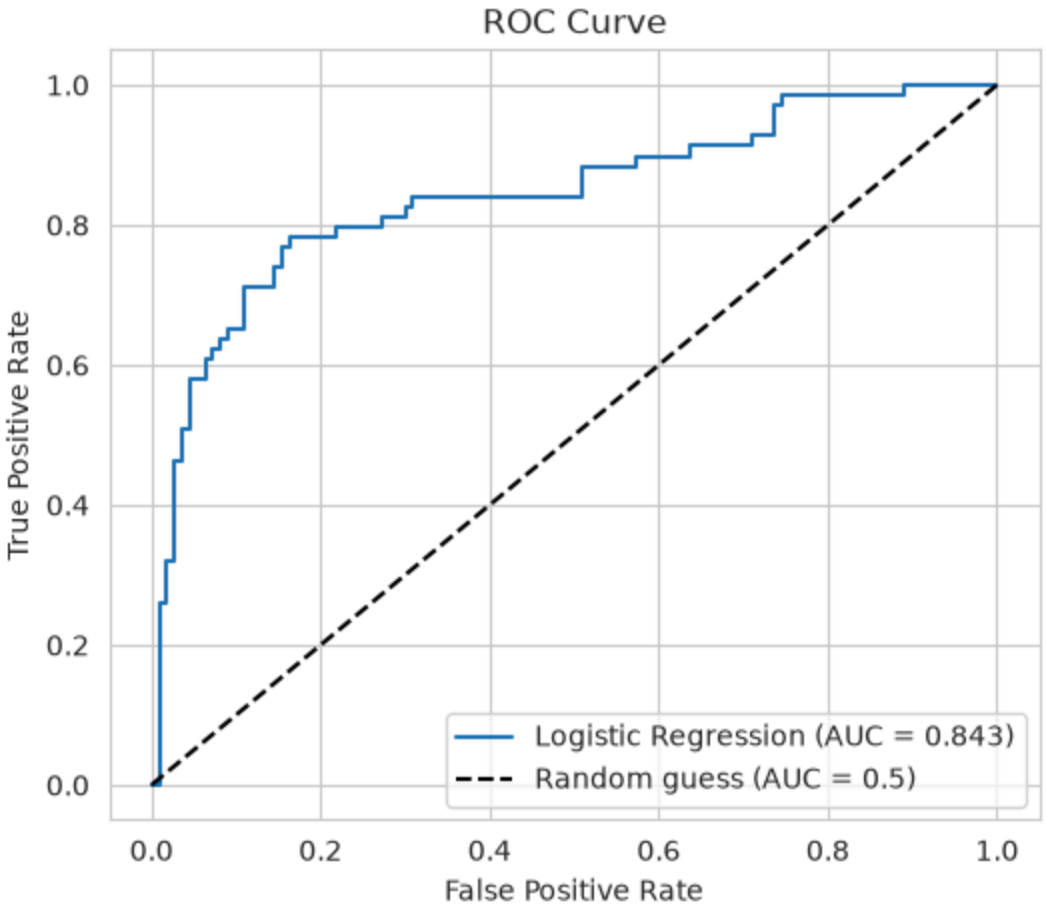
    accuracy                           0.80       179
   macro avg       0.80      0.78      0.79       179
weighted avg       0.80      0.80      0.80       179
```

Interpretation of metrics:

- **Accuracy = 80.4%**: the model correctly predicts 144 out of 179 test passengers. Training accuracy (80.8%) is almost the same, so the model is not overfitting.
- **Precision = 0.79**: when the model says a passenger survived, it is right 79% of the time.
- **Recall = 0.67**: the model finds 67% of the actual survivors; it misses about one third of them.
- **F1-score = 0.72**: balance of precision and recall for the survived class.
- **ROC-AUC = 0.84**: there is an 84% chance the model gives a higher survival probability to a random survivor than to a random non-survivor, which is good separation.



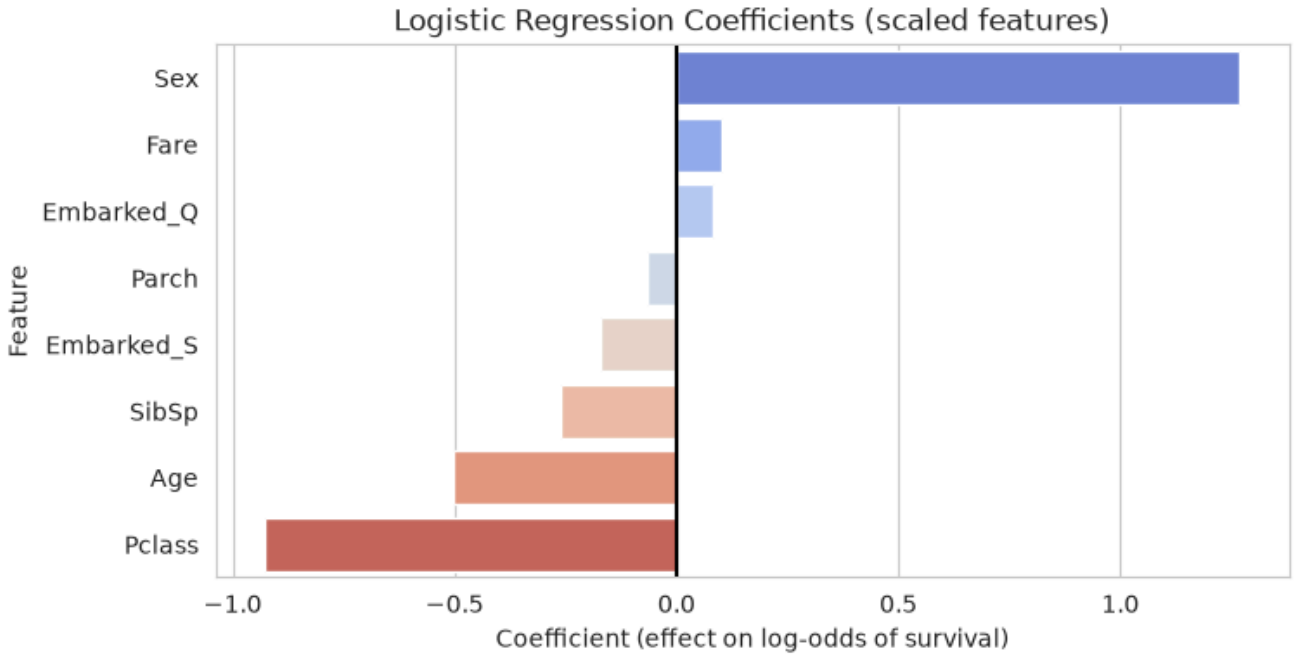
Interpretation: 98 deaths and 46 survivals were predicted correctly. 12 passengers were wrongly predicted as survived (false positives) and 23 survivors were predicted as died (false negatives). The model is better at identifying deaths than survivals, which is why recall for survivors is lower.



Interpretation: The ROC curve stays well above the diagonal line (random guessing) for all thresholds. AUC = 0.84 means the model separates survivors and non-survivors well. If catching more survivors is important, the threshold can be lowered below 0.5 to raise recall at the cost of some precision.

5. Interpretation of Coefficients

Feature	Coefficient	Odds Ratio
Sex	1.269598	3.559422
Fare	0.099599	1.104728
Embarked_Q	0.080683	1.084027
Parch	-0.067158	0.935048
Embarked_S	-0.171784	0.842161
SibSp	-0.262672	0.768994
Age	-0.502878	0.604788
Pclass	-0.928698	0.395068



Interpretation (features are standardised, so coefficients can be compared):

- **Sex (+1.27, odds ratio 3.56):** the strongest factor. Being female increases the odds of survival about 3.6 times per standard deviation, matching the 74% vs 19% survival seen in EDA.
- **Pclass (-0.93, OR 0.40):** second most important. Moving towards 3rd class strongly lowers the odds of survival.
- **Age (-0.50, OR 0.60):** older passengers had lower odds of survival; children were given priority.
- **SibSp (-0.26):** passengers with many siblings/spouses aboard survived less, probably because large families found it harder to escape together.
- **Embarked_S (-0.17):** boarding at Southampton slightly lowers the odds compared to Cherbourg.
- **Fare (+0.10), Embarked_Q (+0.08), Parch (-0.07):** very small effects. Fare is weak here because most of its information is already captured by Pclass.

6. Saving the Model for Streamlit Deployment

The trained model and scaler are saved with pickle so that `app.py` (Streamlit app) can load them and predict for user inputs. Run with `streamlit run app.py`.

```
Saved model.pkl
```

7. Conclusion

- Logistic regression predicted Titanic survival with **80.4% accuracy** and **ROC-AUC of 0.84** on unseen test data, with no sign of overfitting.
- **Sex, passenger class and age** are the most significant predictors. Females, 1st-class passengers and children had the highest chance of survival.

- Recall for survivors (0.67) is lower than precision (0.79), so the model misses some survivors; adjusting the threshold or adding features such as title from Name or family size could improve it.
- The trained model is saved and deployed through a Streamlit app (`app.py`) where a user can enter passenger details and get a survival probability.